NOOPUR TRIVEDI

AI Evaluation & Decision-Governance Researcher (independent) · Senior Business Analyst, Enterprise Transformation

Canada (Canadian citizen) · noopur.trivedi@hotmail.com · linkedin.com/in/noopur-trivedi · github.com/Noopurtrivedi · noopurtrivedi.com

PROFILE

Thirteen years translating messy operational and regulatory problems into structured requirements, test strategies, and governance frameworks, now applied to a hands-on research practice on whether AI systems should be trusted: calibrated decision policies, contamination-controlled faithfulness evaluation, abstention behaviour, and hard-to-game expert rubrics. I build and run my own evaluation pipelines in Python and TypeScript, working daily with Claude Code, local open-weight models, and LoRA fine-tuning. Strongest where expert judgment, risk, and reversibility have to be made measurable.

INDEPENDENT AI RESEARCH & PRODUCT PRACTICE · 2025 – PRESENT

Kestra DML — Decision Management & Governance Layer (Product Architect & Research Lead). A model-agnostic control layer that decides whether, how, and how carefully an LLM or agent should act on a request before any output is generated.

- Authored the taxonomy, decision-metadata schema, benchmark spec, and evaluation framework: six scored dimensions (task type, risk, reversibility, clarification, verification, release action) with asymmetric penalties so under-protection costs double over-protection, plus hard gates for danger-quadrant recall and under-protection rate.
- Hand-labelled a frozen 50-prompt gold dataset; the deterministic v1 engine scores 100% exact match and 100% danger recall on it.
- Designed and ran a 150-case noisy and adversarial benchmark (ambiguous, emotionally framed, under-specified, and manipulative requests) with a written failure analysis, deliberately separated from rule refinement to keep findings honest.
- Specified a 1,000-example corpus strategy with silver/gold tiers, human-review queues, and decision logging.

AdaptIQ — AI Verification & Model-Improvement Layer (Product Architect & Research Lead). An epistemic control layer that makes a base model behave like a careful specialist: grounded, calibrated, self-checking, and willing to abstain.

- Defined the evaluation architecture, verification gates, and a domain contamination rubric (unsupported-claim detection) for finance, legal, and HR tracks; calibrated an LLM judge against human-reviewed gold data.
- Finance evaluation track: reduced contamination from roughly 52% to 17% of answers while raising mean faithfulness from about 0.75 to 0.91, with zero leaked answers on refusal cases and zero false abstentions, across nine verification passes.
- Hands-on with local LLMs (Ollama, MLX on Apple silicon), LoRA fine-tuning, judge frameworks, and compare-arm experiments; documented why tuning moves format and caution but not factual accuracy.

TaxPal — AI-assisted Canadian tax estimation (sole builder). Django/DRF rules engine covering federal and all 13 provincial jurisdictions with every dollar traced to a line item, wrapped in an AI safety layer: immutable system prompt, consent gate, PII redaction, fraud-refusal classifier, risk assessment routing high-risk cases to professional review, append-only audit logs.

Artha — local-first AI agent desktop app (Shree Labs Inc.). 250+ commits across an Electron/TypeScript agent with tool budgets, escalation rules, reversible-plan execution, MCP tools, and Docker-sandboxed execution; no data leaves the machine.

Other shipped work: TrinityTalent career platform (trinitytalent.ai, 700+ commits), Ask Madhav (public, github.com/Noopurtrivedi/ask-madhav), a Claude-powered Google Chat bot on the Vercel Chat SDK and Anthropic SDK, and this portfolio site. All built end to end with Claude Code.

PROFESSIONAL EXPERIENCE

MTU Maintenance Canada — Business Analyst & Project Lead Delta, BC · Apr 2023 – Present

Global aerospace maintenance, repair, and overhaul organization.

- Lead the Canadian workstream of a global data migration and Microsoft 365 modernization program from discovery through cutover, validation, and stabilization.
- Developed governance for metadata, permissions, retention, export compliance, and sensitive HR/HSE data, including escalation to leadership and legal consultation on cloud-migration risk decisions.
- Directed requirements engineering for an enterprise LMS replacement: 160+ functional, security, integration, reporting, and operational requirements with traceability.
- Designed UiPath automation moving SAP export data into the quality issue-management workflow; led legacy Access modernization analysis toward SQL Server and Dataverse.
- Identified adoption gaps and built a Microsoft 365 enablement program: 30+ sessions, ~250 users trained directly, training coordinated for 700+ employees.

BC Hydro — SAP Business Decision Analyst (Consultant) Burnaby, BC · Oct 2022 – Oct 2023

- Translated business, operational, and technology needs into structured requirements, acceptance criteria, and governance artifacts for SAP EAM and CCS initiatives at a provincial Crown utility.
- Developed test strategies and traceability documentation connecting business needs with implementation teams.

Bell Canada / Bell Media — Business Intelligence Developer (Consultant) Toronto, ON · Nov 2021 – Apr 2023

- Built BI infrastructure, ETL workflows, SQL, and Pentaho/CTools executive reporting for a media marketplace; resolved data-quality issues and trained users on self-service analytics.

NRI Distribution — Systems Integration Specialist Calgary, AB · Aug 2021 – Feb 2023

- Designed API and EDI integrations between internal systems and customer ERPs; traced and resolved a critical EDI label-pricing defect end to end.

X360 Digital — Quality Analyst Remote, Canada · Mar 2019 – Dec 2021

- Automated QA and data-reporting pipelines (SQL, Excel); executed 200+ tests, contributing to a 31% reduction in defect rates.

Syncrude Canada · Opal Marketing Group · Acrodex / Imperial Oil AB & BC · 2012 – 2019

- Planning and cost analysis, CDMS administration, project coordination, and technical analysis in oil-sands operations.

TECHNICAL SKILLS

Research & evaluation: benchmark and gold-dataset design, expert rubrics, LLM-as-judge calibration, contamination and faithfulness metrics, abstention and refusal analysis, adversarial test sets, failure analysis.
AI engineering: Python, TypeScript, Claude Code and Claude API, Anthropic SDK, Ollama, MLX, LoRA/QLoRA, RAG, MCP tools, agent orchestration, Docker sandboxing.
Enterprise: SQL, Power BI, Power Query, SAP, Microsoft 365/Azure, UiPath, requirements engineering, test strategy, information governance.

EDUCATION & CERTIFICATIONS

Bachelor of Business Administration, Marketing — Thompson Rivers University, Kamloops, BC, 2013

Certified Scrum Master (Scrum Alliance, 2021) · Azure AI Fundamentals (Microsoft, 2021) · ISTQB Certified Tester Foundation Level (ATSQ, 2021) · Software Product Management (University of Alberta, 2020) · Full Stack Development (EvolveU, 2020)